

Understanding the rules score

The Rules tab on the Findings page displays a rules score, which helps indicate the severity of the rule breaks. You can configure the methodology used to calculate the rules score on the Rule Details dialog.

Note The rules score is entirely separate from the data quality score. For more information on the data quality score, go to [Data Quality Score Meter](#).

Rules settings

The Rule Details dialog provides the following options for calculating the rules score.

The screenshot shows the Rule Details dialog with the following settings:

- Score:** Three tabs labeled 'Low', 'Medium', and 'High'. The 'Low' tab is selected. An information icon (i) is to the right.
- Scoring Type:** A dropdown menu set to 'Percent'.
- Points *:** A text input field containing the value '2', with a green checkmark to its right.
- Percent *:** A text input field containing the value '5', with a green checkmark to its right.
- Tolerance (i):** A text input field containing the value '0'.

Score

The Score widget provides three options, or presets, for calculating the rules score. Select an option to automatically populate the Points field and Percentage field (if applicable) with predefined values. You may modify the values as needed.

- | | |
|---------------|---|
| Low | Downscores 1 point for every 1% of records that violate a rule. For example, if 25 of 100 records break a rule, 25 points are deducted from the rules score. |
| Medium | Downscores 5 points for every 1% of records that violate a rule. For example, if 2 of 100 records break a rule, then 10 points are deducted from the rules score. |

High Downscores 25 points for every 1% of records that violate a rule. For example, if 3 of 100 records break a rule, 75 points are deducted from the rules score.

Note When you manually update the score, the button automatically adjusts to match the corresponding scoring range:

- **Low:** 1-4 points
- **Medium:** 5-24
- **High:** 25-100

Scoring type

The Scoring Type field determines the methodology used to calculate the rules score.

Percent Deducts points based on the percentage of records that violate the rule. Use the **Points** and **Percent** fields to set the percentage threshold. For example, suppose you have 100 records, and the **Points** and **Percent** fields are both set to 1. If 25 of the records break your rule (which is 25%), then a total of 25 points are deducted from your rules score.

Absolute Deducts a fixed number of points when one or more rules break, regardless of the percentage of records that break the rule. For example, if you set the value of the **Points** field to 10 and a rule breaks, a total of 10 points are deducted from your rules score.

The Absolute option is useful when you want to know if any rule breaks occur in the findings. Note that if more than one rule breaks, the point deduction is the same as if only one rule breaks.

Points

Used with both the Percent and Absolute scoring types. Set Points to a value between 0 and 100, which is the number of points deducted from the rules score when a record violates a rule.

Percent

Used with the Percent scoring type. Set Percent to a value between 0 and 100, which is the ratio of the number of breaking records over the total number of records.

Tolerance

Specifies the tolerance threshold as a percentage. Enter a percentage that represents the acceptable amount of rule breaks. For example, if the Tolerance is set to 10%, and the percentage of rule breaks is less than or equal to 10%, such as 5%, the rule status is passing and the score is unaffected. If the percentage of rule breaks is greater than 10%, such as 15%, the rules score is calculated based on the settings provided.

Example of the Percent scoring method

In this example, you have 100 records and you would like to deduct 2 points for every 5% of records that break the rule. Set the following parameters:

- Scoring Type: Percent
- Points: 2
- Percent: 5

Score	Scoring Type
Low Medium High ⓘ	Percent ▼
Points *	Percent *
2 ✓	5 ✓
Tolerance ⓘ	
0	

Example results are:

- If 3 records break the rule, 0 points are deducted from the rules score.
 - 3% of the records break (3/100)
 - 3% is less than or equal to the 5% threshold.
- If 15 records break the rule, 6 points are deducted from the rules score.
 - 15% of the records break (15/100)
 - 2 points are deducted per 5% of rule breaks (2 points * (15% / 5%) = 6).
- If 25 records break the rule, 10 points are deducted from the rules score.
 - 25% of the records break (25/100)
 - 2 points are deducted per 5% of rule breaks (2 points * (25% / 5%) = 10).

Rules (Score: 16)										
Rule Name	Rule Query	Points	Tolerance	Perc	Breaking Records	Passing Records	Total Records	Status	Dimension	State
limit15	SELECT * FROM berka_accounts_17_dataset limit 15	6	N/A	15.0000%	15	85	100	BREAKING		Enabled
limit25	SELECT * FROM berka_accounts_17_dataset limit 25	10	N/A	25%	25	75	100	BREAKING		Enabled
limit3	SELECT * FROM berka_accounts_17_dataset limit 3	0	N/A	3.0000%	3	97	100	BREAKING		Enabled

Now let's change the value in the **Tolerance** field from 0 to 15%, indicating the acceptable amount of rule breaks. The results are:

- If 3 records break the rule, it is below the Tolerance of 15%. 0 points are deducted from the rules score. An asterisk (0*) indicates that rule breaks occurred, but they did not affect the rule score.
- If 15 records break the rule, it is equal to the Tolerance of 15%. 6 points are deducted from the rules score.
- If 25 records break the rule, it is above the Tolerance of 15%. 10 points are deducted from the rules score.

Rules (Score: 16)										
Rule Name	Rule Query	Points	Tolerance	Perc	Breaking Records	Passing Records	Total Records	Status	Dimension	State
limit15	SELECT * FROM berka_accounts_17_dataset limit 15	6	15%	15.0000%	15	85	100	BREAKING		Enabled
limit25	SELECT * FROM berka_accounts_17_dataset limit 25	10	15%	25%	25	75	100	BREAKING		Enabled
limit3	SELECT * FROM berka_accounts_17_dataset limit 3	0*	15%	3.0000%	3	97	100	PASSING		Enabled

Note If the number of rule breaks is above the Percent value but below the Tolerance value, 0 points are deducted from the rules score. The asterisk (0*) indicates that rule breaks occurred, but they did not affect the rule score.

Example of the Absolute scoring method

In this example, you have 100 records and you would like to deduct 2 points whenever a rule is broken, regardless of the number of records breaking the rule. Set the following parameters:

- Scoring Type: Absolute
- Points: 2

Score

Low

Medium

High

i

Scoring Type

Absolute

▼

Points *

2

✓

Tolerance i

0

Example results are:

- If 3 records break the rule, 2 points are deducted from the rules score.
- If 15 records break the rule, 2 points are deducted from the rules score.
- If 25 records break the rule, 2 points are deducted from the rules score.

Rules (Score: 6) i											
Behavior Rules (Score: 6) i Outliers Pattern Source Record Schema Dupes Shapes											
Labels Job Export											
Rule Discovery Runtime: 00:00:03											
Rule Name ↑	Rule Query	Points i	Tolerance i	Perc i	Breaking Records	Passing Records	Total Records i	Status i	Dimension	State	Status i
+ Limit15	SELECT * FROM temp_3743780826752_berka_accounts_17_dataset limit 15	2	N/A	15.0000%	15	85	100	BREAKING		Enabled	Unassigned
+ Limit25	SELECT * FROM temp_3743780826752_berka_accounts_17_dataset limit 25	2	N/A	25%	25	75	100	BREAKING		Enabled	Unassigned
+ Limit3	SELECT * FROM temp_3743780826752_berka_accounts_17_dataset limit 3	2	N/A	3.0000%	3	97	100	BREAKING		Enabled	Unassigned

The Absolute method ensures that a consistent penalty is applied each time the rule is violated, regardless of the extent of the violation.

Now let's change the value in the **Tolerance** field from 0 to 15%, indicating the acceptable amount of rule breaks. The results are:

- If 3 records break the rule, it is below the Tolerance of 15%. 0 points are deducted from the rules score. An asterisk (0*) indicates that rule breaks occurred, but they did not affect the rule score.
- If 15 records break the rule, it is equal to the Tolerance of 15%. 2 points are deducted from the rules score.
- If 25 records break the rule, it is above the Tolerance of 15%. 2 points are deducted from the rules score.

Behavior Rules (Score 4) Outliers Pattern Source Record Schema Dupes Shapes														Labels Job Export		
														Rule Discovery Runtime: 00:00:02		
Rule Name ↑	Rule Query	Points ⓘ	Tolerance ⓘ	Perc ⓘ	Breaking Records	Passing Records	Total Records ⓘ	Status ⓘ	Dimension	State	Status ⓘ	Profile	Actions ⓘ			
+ Limit15	SELECT * FROM temp_1743780826752_berka_accounts_17_dataset limit 15	2	15%	15.0000%	15	85	100	BREAKING		Enabled	Unassigned		Actions			
+ Limit25	SELECT * FROM temp_1743780826752_berka_accounts_17_dataset limit 25	2	15%	25%	25	75	100	BREAKING		Enabled	Unassigned		Actions			
+ Limit3	SELECT * FROM temp_1743780826752_berka_accounts_17_dataset limit 3	0*	15%	3.0000%	3	97	100	PASSING		Enabled	Unassigned		Actions			

Add CA certificates to the java keystore

This topic describes how to add CA certificates to the java keystore in a Kubernetes deployment. The following approach mounts the directory containing the SSL certificates from the Agent pod as a shared volume. The SSL certificates are then accessible to the Spark driver/executor pods, allowing them to establish a secure SSL/TLS connection with the email server and/or data source.

Steps

1. Mount the directory containing the SSL certificates in the Agent pod to a shared volume. This can be done in the deployment or pod configuration file for the Agent pod in Kubernetes. For example, you can add a volume and volumeMounts section to the Agent pod's configuration file like this:

```
volumes:
  - name: certs
    hostPath:
      path: /etc/pki/ca-trust/extracted/java/cacerts
```

2. Modify the Spark driver/executor pod's configuration to mount the shared volume containing the SSL certificates from the Agent pod. This can be done in the Spark configuration or pod configuration file. For example, you can add a volume and volumeMounts section to the Spark driver/executor pod's configuration file like this:

```
volumes:
  - name: certs
    hostPath:
      path: /path/to/agent/certs
```

3. Update the Java options for the Spark driver/executor pods to specify the path to the SSL certificates in the shared volume. You can do this by adding the `-Djavax.net.ssl.trustStore` and `-Djavax.net.ssl.trustStorePassword` options to the Spark driver/executor pod's configuration file, like this:

```
javaOptions:
  - "-Djavax.net.ssl.trustStore-
e=/path/to/agent/certs/cacerts"
  - "-Djavax.net.ssl.trustStorePassword=your_keystore_pass-
word"
```

4. Replace `/path/to/agent/certs` with the actual path to the shared volume where the SSL certificates are mounted in the Spark driver/executor pods, and `your_key-store_password` with the actual password for the keystore that contains the SSL certificates.
5. Restart the Spark driver/executor pods to apply the changes.

Migrating keystores

This section describes the process for rotating or migrating encryption keys used to encrypt data at rest in the PostgreSQL metastore. The process for rotating keys includes the following steps:

- Create your keystore. The keystore must be in one of the following formats: BCFKS, JCEKS, or PKCS12. Note that BCFKS is required for [FIPS compliance](#). Both the keystore and the private key must be password-protected. The keystore file must be located in a path accessible by the running application. For more information on creating an initial keystore, go to [Setting up a custom keystore](#).
- Choose a method for providing secrets:
 - Environment variables (recommended for self-hosted Kubernetes): The best practice for cloud-native and containerized environments. For this, simply set the relevant environment variables.
 - File-based configuration tree (recommended for Standalone servers): A secure alternative for single-server or non-containerized deployments or an alternative approach for containerized deployments.
- Set the properties. The following sections describe the necessary properties.

Set migration properties

Declare the following migration properties.

Environment variable	Description
DQ_APP_ENCRYPTION_AT_REST_MODE	Specifies whether to use the legacy or hybrid cryptosystem. To migrate keystores, set to hybrid . Default is legacy .
DQ_APP_ENCRYPTION_AT_REST_MIGRATION	For existing customers that need to migrate from legacy to hybrid encryption at rest mode. When set to true , performs migration to new encryption key during startup. After completing the initial migration, disable this setting. Default is false .

Set new keystore properties

Declare the following information about the new keystore.

Environment variable	Description
KEYSTORE_SECRETS_SOURCE	The type of keystore secrets source. For example, file .
KEYSTORE_FILEPATH	The path to the new keystore file. For example, /app/config/new-keystore.jceks .
KEYSTORE_TYPE	The type of the new keystore file. Allowable values are BCFKS , PKCS12 (extension .p12) or JCEKS .
KEYSTORE_PASSWORD	The password for the new keystore file. We recommend using a configuration tree file for this value.
KEYSTORE_PRIVATEKEY_PASSWORD	The password for the new private key. We recommend using a configuration tree file for this value.
	Note Because of a standard java implementation detail, for PKCS12 format only, the private key password and the keystore pass often need to be the same value.
KEYSTORE_PRIVATEKEY_ALIAS	The alias for the new private key.
KEYSTORE_PUBLICKEY_ALIAS	The alias for the new public key.

Set prior keystore properties

If migrating to a new keystore, you must declare information about the prior keystore.

Environment variable	Description
KEYSTORE_ PRIOR_ SECRETS_ SOURCE	The type of the prior keystore secrets source. For example, file .
KEYSTORE_ PRIOR_ FILEPATH	The path to the old (prior) keystore file. For example, /app/config/old-keystore.jceks .
KEYSTORE_ PRIOR_ TYPE	The type of the prior keystore file. Allowable values are BCFKS , PKCS12 , or JCEKS .
KEYSTORE_ PRIOR_ PASSWORD	The password for the prior keystore file. We recommend using a configuration tree file for this value.
KEYSTORE_ PRIOR_ PRIVATEKEY_ PASSWORD	The password for the prior private key.
KEYSTORE_ PRIOR_ PRIVATEKEY_ ALIAS	The alias for the prior private key.

Secure the master password

For enhanced security, we recommend using a file-based configuration tree to store critical secrets like passwords. This is a powerful feature of Spring Boot. For more details, go to the [Spring Boot documentation](#) on external configuration.

To use this method, specify the path to your secret directory in your primary configuration file, `application.properties`.

```
spring.-  
config.import=optional:configtree:/opt/owl/config/secrets/
```

The passwords will be read from files with the same name as their keys. For example, the contents of a file at the location `keystore/privatekey/password` relative to the `spring.config.import` path will be used for the `KEYSTORE_PRIVATEKEY_PASSWORD` variable.

Perform the migration

After setting the properties, start your DQ Application. The migration runner automatically detects the configuration and executes the migration scripts. Once the migration is complete, remove the migration properties and the prior keystore properties to return the application to its normal operational state.

Note If multi-tenant mode is enabled, note that Data Quality & Observability Classic supports only one keystore for the entire installation, across all tenants. Accordingly, the migration for multi-tenant mode applies to all tenants, both enabled and disabled tenants will be migrated.

Multi-tenancy and Tenant Manager

Collibra DQ supports multi-tenant environments. Customers can use the Tenant Manager application to manage their tenants.

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Enabling multi-tenancy

Multi-tenancy provides the ability to instantiate different organizations within one entity. For example, an organization could have two separate divisions called OrgTraders and OrgInsurance, respectively. The OrgTraders division is not allowed to see the DQ Check results of the OrgInsurance division, making it critical that these organizations remain separate in Collibra DQ. With multi-tenancy, you can segregate these organizations into two separate tenants within the overarching Collibra DQ web application to protect the activity of one organization from the other.

After establishing the multi-tenant environment, use the Tenant Manager application to manage the tenants. For more information, go to [Using Tenant Manager](#). Note that the Tenant Manager application is available only after enabling multi-tenancy.

Prerequisites

Non-URL-based tenancy

- DNS entry for owl web server IP.
- DNS entry for multiTenantSchemaHub.

URL-based tenancy: tenant.host

- DNS entry for each tenant.
- DNS entry for every tenant you want to create (the example below is called "automation").

Note Each of the records above point to the same IP address.

Steps

Standalone

Note If you are upgrading your version of Collibra DQ, go to [Upgrade Data Quality & Observability Classic](#).

1. Make sure the web application has started up one time and that you have successfully logged into it with the default credentials.
2. Stop all the components using the `./owlmanage.sh stop` command.
3. Modify the `owl-env.sh` file to include the following parameters:
 - `export multiTenantSchemaHub=owlhub`: This is a new schema that will get created on owlweb start (note the name of the TenantSchemaHub can be changed to the desired name at setup time).
 - `export MULTITENANTMODE=TRUE`: This enables multi-tenancy mode.
 - `export URLBASEDMULTITENANTMODE=TRUE/FALSE`
 - **TRUE (default)**: This value means that you are using the tenant name as a sub-domain (see prerequisites).
 - **FALSE**: This value means you will let Collibra DQ manage tenants via sessions/tokens.
4. If using agents as part of the DQ operation, modify the `owl.properties` file to include the following:
 - `spring.agent.datasource.url=jdbc:postgresql://cdh-edge-dan.us-east4-c.c.owlhadoop-cdh.internal:5432/postgres**?currentSchema=owlhub**` (matching the name of the schema set on step 3-1 above).

- jdbc:postgresql://cdh-edge-dan.us-east4-c.c.owl-hadoop-cdh.internal:5432/postgres**?currentSchema=owlhub** (matching the name of the schema set on step 3-1 above).
- 5. Once the settings have been configured for multi-tenancy, start up the owlweb host first, using `./owlmanage.sh start=owlweb` command.
- 6. Start up the agents, using the `./owlmanage.sh start=owlagent` command.

Note To use multi-tenancy when `URLBASEDMULTITENANTMODE=TRUE`, make sure there are DNS entries to the tenant endpoints.

Kubernetes

You can enable multi-tenancy in the web ConfigMap for Kubernetes. The high-level setup for Kubernetes is virtually the same as standalone deployments. Where you add or modify properties in `owl-env.sh` for standalone deployments, you add the properties to the ConfigMap for Kubernetes deployments.

If you are using Helm charts, you can customize and modify configuration files to suit the needs of different environments. The Helm charts define all sets of properties.

The following is an example of the type of configurations you might see when running the Helm chart to deploy Kubernetes. These configurations are located in the `values.yaml` file.

```
flyway: false
server_port: 9005
server_port_2: 9001
multitenant_mode: false
multitenant_schema_hub: owlhub
url_based_multitenantmode: false
owl_base: /opt
owl_agent_id: 0
install_path: /opt/owl
spark_home: /opt/spark
```

For more information about using Helm charts, go to **Collibra DQ Helm chart** in [Cloud native](#).

What's next

[Using Tenant Manager](#)

Configuring SSO for Tenant Manager

Using Tenant Manager

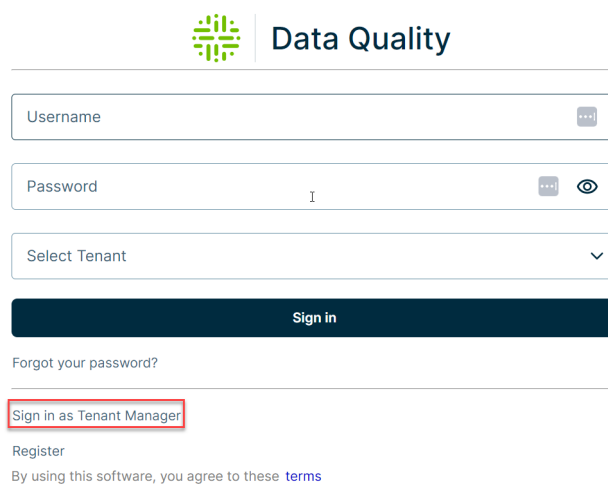
Tenant Manager is a separate DQ application that enables you to manage [multiple tenants](#). You can also configure single sign-on (SSO) and Active Directory security for Tenant Manager.

Logging in to Tenant Manager

Tenant Manager is available only after [enabling multi-tenancy](#), and it requires a separate login.

Steps

1. Sign into Collibra DQ.
2. Click **Sign in as Tenant Manager**, which appears below the **Forgot your password?** link.



 Data Quality

Username 

Password  

Select Tenant 

Sign in

[Forgot your password?](#)

Sign in as Tenant Manager

[Register](#)

By using this software, you agree to these [terms](#)

3. On the Tenant Manager login page, enter the username and password provided during your service engagement.

Note You can change the password after logging in to Tenant Manager. Click the user profile icon in the upper right corner and select **Change Password**.



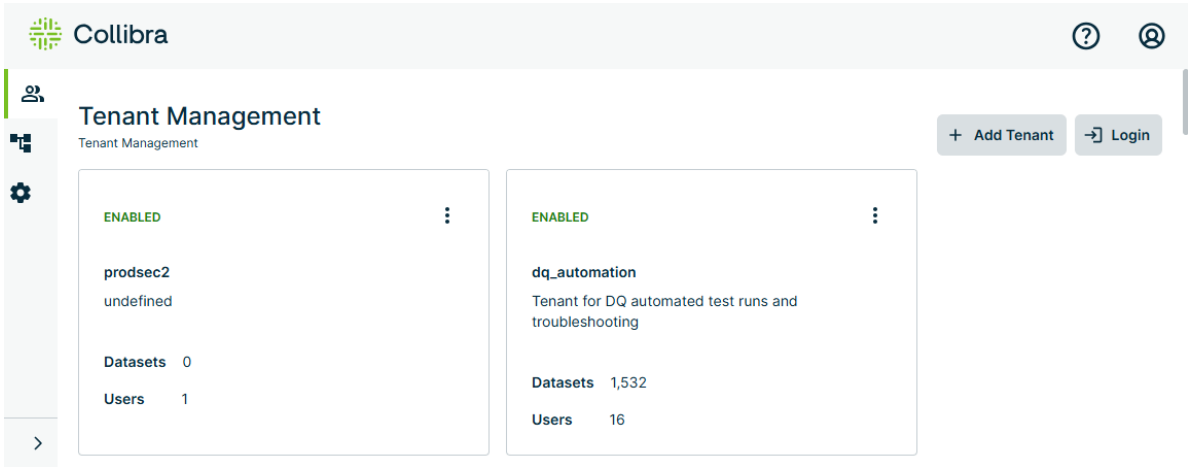
Data Quality Tenant Manager

[Log in](#)

[Forgot Password?](#)

4. Click **Log in**.

» The Tenant Management page opens, where you can add, edit, and delete your tenants.



Collibra

Tenant Management

[+ Add Tenant](#) [Login](#)

Tenant Name	Status	Datasets	Users
prodsec2	ENABLED	0	1
dq_automation	ENABLED	1,532	16

Adding a tenant

You can add a tenant from the Tenant Management page.

Steps

1. Click the **+ Add Tenant** button in the top-right corner of the Tenant Management window.
 - » The Add a New Tenant modal appears.

Add a New Tenant ×

☒ Enable

Tenant Id(Unique)

Tenant Name: automation

Tenant Display Name: automation

Description: Automated testing tenant

Tenant Credentials

Username *: automation

Password *: Mtadmin123\$

Confirm Password *:

Cancel Save

2. Enter the following information:
 - **Tenant Name:** Name for the tenant.
 - **Tenant Display Name:** Tenant name that displays in the Tenant Management page.
 - **Description:** Description for the tenant.
 - **Username:** Username you use when you login to the tenant. This is not required if the tenant is SAML-enabled.

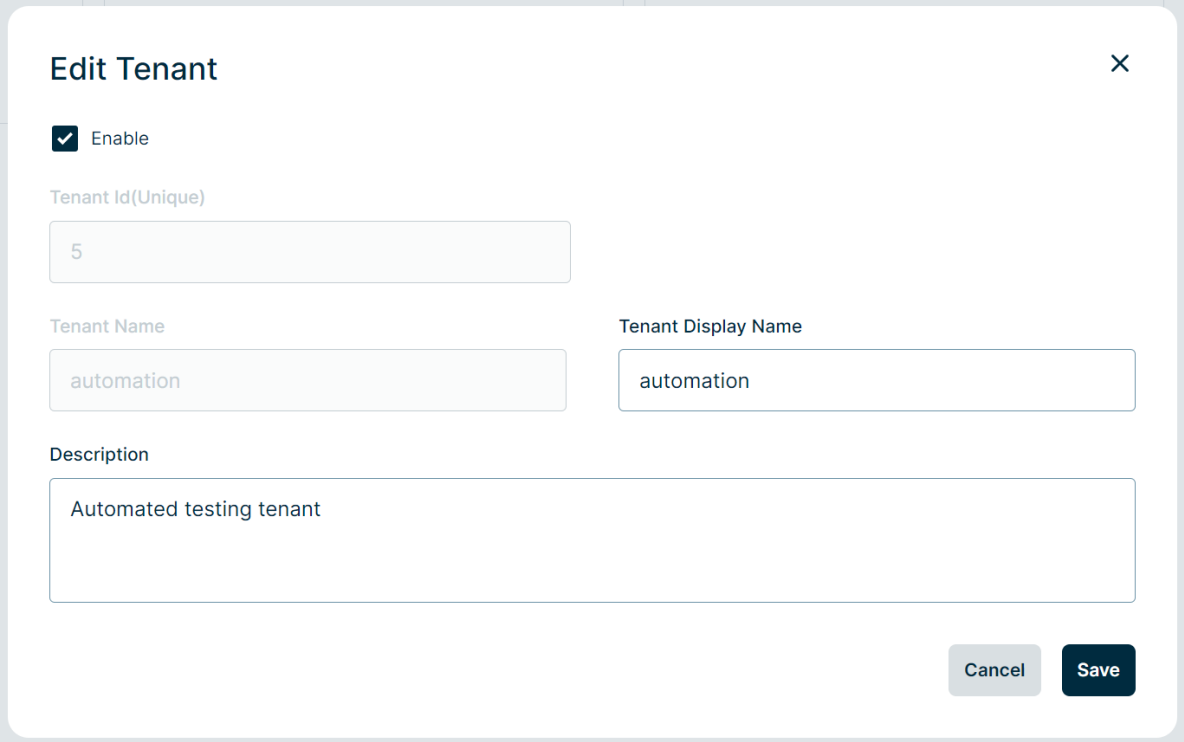
- **Password:** Password you use when you login to the tenant. This is not required if the tenant is SAML-enabled.
- 3. To enable the tenant, select the **Enable** box.
- 4. Click **Save**.
 - » The new tenant displays on the Tenant Management page.

Editing a tenant

You can edit a tenant from the Tenant Management page.

Steps

1. Select the tenant from the Tenant Management page and click the icon next to the **Login** button.
2. Select **Edit** from the drop-down.
 - » The Edit Tenant modal appears.



The screenshot shows a modal window titled "Edit Tenant" with a close button (X) in the top right corner. Inside the modal, there is a checkbox labeled "Enable" which is checked. Below this, there are three input fields: "Tenant Id(Unique)" containing the value "5", "Tenant Name" containing the value "automation", and "Tenant Display Name" containing the value "automation". At the bottom, there is a "Description" field containing the text "Automated testing tenant". In the bottom right corner of the modal, there are two buttons: "Cancel" and "Save".

3. Make the desired changes and click **Save**.

Deleting a tenant

You can delete a tenant from the Tenant Management page.

Steps

To delete a tenant, on the Tenant Management page, click the icon and select **Delete**.

Configuring SAML security settings

You can add, edit, or delete a tenant's SAML security settings from the Tenant Management page.

Steps

To configure a tenant's SAML security settings, on the Tenant Management page, click the icon and select **SAML Config**. See [SAML Security Settings Overview](#) for more information on configuring the settings.

Once the tenant administrator configures the security settings for the SAML page and updates the configuration with `SAML_TENANT_MANAGER_ENABLED=true`, credentials are no longer required to add a new tenant.

Mapping roles to tenants

You can map roles to tenants from the Tenant Management page.

Steps

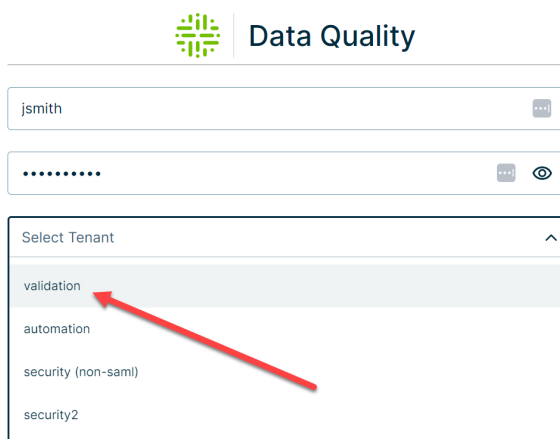
To map roles to tenants, on the Tenant Management page, click the icon and select **Role Mapping**. See [AD Group to Role Mapping](#) for more information on mapping roles.

Signing into a tenant

You can select a tenant from a drop-down list on the main Collibra DQ login page.

Steps

1. Sign into Collibra DQ.
2. Enter your username and password credentials.
3. Select your tenant from the Select Tenant drop-down list, which lists the tenants in alphabetical order.



The screenshot displays the Collibra Data Quality login page. At the top, there is a green Collibra logo followed by the text 'Data Quality'. Below this, there are two input fields: the first for the username, which contains 'jsmith', and the second for the password, which is masked with dots. Below the password field is a 'Select Tenant' dropdown menu. The dropdown is open, showing a list of tenants: 'validation', 'automation', 'security (non-saml)', and 'security2'. A red arrow points to the 'validation' tenant, indicating it is the selected option.

Note All enabled tenants will display in the Select Tenant drop down list. Access to tenants are handled by the administrator(s) within each tenant individually.

4. Click **Sign in**.

Note If you haven't registered as a Collibra DQ user, click the **Register** link, which displays under the **Sign in as Tenant Manager** link and complete the registration form.

Configuring security settings

Tenant Manager supports the following security options:

- Single-sign-on (SSO). For more information, go to [Configuring SSO for Tenant Manager](#)
- Active Directory. The Active Directory configuration options for Tenant Manager are identical to the settings provided for the DQ Application. For more information, go to [AD and LDAP Security Settings Overview](#).

Configuring SSO for Tenant Manager

Tenant Manager is a separate DQ application for managing multi-tenant environments. The following steps describe how to enable single sign-on (SSO) for the Tenant Manager. For more information on Tenant Manager, see [Using Tenant Manager](#).

Important The Tenant Manager is considered a new application by the entity provider (Idp). Do not change any existing IdP or SSO configuration settings for the DQ Web application.

Configure the identity provider for the new application

Begin by configuring the IdP for the Tenant Manager application.

1. Set the **Identifier (Entity ID)** for the new IdP application as `DQTenantManager`. Do not change the existing entity ID for the original SSO application.
2. Set the **Reply URL (Assertion Consumer Service URL)** for the new IdP application to the following (replacing `{host}` with your DQ host endpoint):
`https://{host}/tenant/saml/sso/tenantmanager`
3. Assign tenant admin users to the IdP application.
4. Assign tenant admin groups to the IdP application.

Set the SAML authentication properties

Add the following properties to your configuration.

Standalone

1. Add the following property to your owl-env.sh file:

```
export SAML_TENANT_MANAGER_ENABLED=true
```
2. Add additional properties as needed (see below). Prefix each property with the export statement.
3. Restart the DQ Web application.

Cloud Native

1. Add the following property as an environment variable to your DQ-web Configmap:

```
SAML_TENANT_MANAGER_ENABLED: true
```
2. Add additional properties as needed (see below)
3. Recycle the pod.

Required properties

Property	Description
SAML_ENTITY_BASEURL	The base URL that is provided in the service provider metadata. It is used to generate the ACS URL, which is the entity base URL with <code>/saml/sso</code> appended to it. It must include a port number if one is configured on the IdP side. Once you have enabled and configured SAML authentication, you can download the service provider metadata from <code>https://<your_dq_environment_url>/saml/metadata/SSO</code>. Important If you use Data Quality & Observability Classic with a Java version older than 17, such as Java 11, download your service provider metadata from <code>https://<your_dq_environment_url>/saml/metadata</code> instead.

Warning If you have SAML configured in Collibra DQ, or if the app sits behind a load balancer, see the `CORS_ALLOWED_ORIGINS` property below.

Optional properties: general

Property	Description
CORS_ALLOWED_ORIGINS=\${IDP-BASE-URL},\${DQ-BASE-URL}	Allows cross-origin requests between Colibra DQ and SAML. Replace <code>\${IDP-BASE-URL}</code> with the value of the actual IdP URL. For example, <code>https://ping.auth.com/</code>. Replace <code>\${DQ-BASE-URL}</code> with the value of the actual DQ Base URL. For example, <code>https://dq-env.com</code>.
SAML_LB_EXISTS	Whether the application needs to configure a load balancer. You generally need this setting only when the Load Balancer is set for SSL Termination. The default value is <code>false</code>. If set to <code>true</code>, you must also provide a value for <code>SAML_LB_SERVER_NAME</code>. Note If the <code>SAML_LB_EXISTS</code> property is set to <code>true</code> and <code>SAML_LB_INCLUDE_PORT_IN_REQUEST</code> set to <code>false</code>, you may need to update <code>SAML_ENTITY_BASEURL</code> to include the port in the URL. The <code>SAML_ENTITY_BASEURL</code> should match the IdP's ACS URL.
SAML_ROLES_PROP_NAME	The attribute in which the identity provider stores the role of the user authenticating in the SAML response. The default value is <code>memberOf</code>.
SAML_USER_NAME_PROP	The name of the attribute in the SAML response that contains the username of the user who is authenticating.
SAML_KEYSTORE_FILE	The path to the keystore for SSL validation. The store should contain the keypair of the identity provider for SSL verification.
SAML_KEYSTORE_PASS	The password for the keystore provided in <code>SAML_KEYSTORE_FILE</code> .

Property	Description												
SAML_ KEYSTORE_ALIAS	The alias of the keypair (private and public) in the keystore used for SSL verification.												
LOCAL_ REGISTRATION_ ENABLED	Allows you to display or hide the registration link for local users on the Collibra DQ sign-in page. Because the registration link is displayed on the sign-in page by default, this property is set to <code>true</code> . To display or hide the registration link from the owl-env.sh file, select from the following properties:												
	<table> <tr> <th>Property</th><th>Description</th></tr> <tr> <td>export LOCAL_ REGISTRATION_ ENABLED=true</td><td>Displays the registration link on the sign-in page for local users. This is set to <code>true</code> by default.</td></tr> <tr> <td>export LOCAL_ REGISTRATION_ ENABLED=false</td><td>Hides the registration link on the sign-in page for local users.</td></tr> </table> To display or hide the registration link from a K8s ConfigMap, select the from the following properties: <table> <tr> <th>Property</th><th>Description</th></tr> <tr> <td>LOCAL_ REGISTRATION_ ENABLED:"true"</td><td>Displays the registration link on the sign-in page for local users. This is set to <code>true</code> by default.</td></tr> <tr> <td>LOCAL_ REGISTRATION_ ENABLED:"false"</td><td>Hides the registration link on the sign-in page for local users.</td></tr> </table>	Property	Description	export LOCAL_ REGISTRATION_ ENABLED=true	Displays the registration link on the sign-in page for local users. This is set to <code>true</code> by default.	export LOCAL_ REGISTRATION_ ENABLED=false	Hides the registration link on the sign-in page for local users.	Property	Description	LOCAL_ REGISTRATION_ ENABLED:"true"	Displays the registration link on the sign-in page for local users. This is set to <code>true</code> by default.	LOCAL_ REGISTRATION_ ENABLED:"false"	Hides the registration link on the sign-in page for local users.
Property	Description												
export LOCAL_ REGISTRATION_ ENABLED=true	Displays the registration link on the sign-in page for local users. This is set to <code>true</code> by default.												
export LOCAL_ REGISTRATION_ ENABLED=false	Hides the registration link on the sign-in page for local users.												
Property	Description												
LOCAL_ REGISTRATION_ ENABLED:"true"	Displays the registration link on the sign-in page for local users. This is set to <code>true</code> by default.												
LOCAL_ REGISTRATION_ ENABLED:"false"	Hides the registration link on the sign-in page for local users.												
Note This property must be set globally on deployment.													
CSRF_TOKEN_ ENABLED	Enables the use of Spring CSRF token implementation.												
CSRF_TOKEN_ IGNORE_URL_ PATTERN	Requests URL patterns to be ignored for CSRF token validation.												

Property	Description
CSRF_TOKEN_IGNORE_REFERER_PATTERN	Requests header referer patterns to be ignored for CSRF token validation.

Note While CORS is still an optional configuration, it is required if you have SAML configured in Collibra DQ, or if you have Collibra DQ behind a load balancer.

CORS is also enforced for multi-tenancy.

Optional properties: load balancer

When `SAML_LB_EXISTS` is set to `true`, the following additional properties are available.

Property	Description
SAML_LB_INCLUDE_PORT_IN_REQUEST	Whether to include the port number in the request. The default value is <code>false</code> .
SAML_LB_PORT	The port number of the load balancer. The default value is <code>443</code> .
SAML_LB_SCHEME	The protocol of the load balancer. The default value is <code>https</code> .
SAML_LB_SERVER_NAME	The server or DNS name. Usually, the same as <code>SAML_ENTITY_BASEURL</code> without specifying the protocol, for example without <code>https://</code> . This property is required and has no default.
SAML_LB_CONTEXT_PATH	Any path that may be defined on the load balancer.

Update settings in Tenant Manager

When you are ready with your IdP settings, add the final configuration settings in the Tenant Manager:

1. Sign in to Tenant Manager using the **Sign in as Tenant Manager** link.
2. Click in the left navigation pane.
 - » The SAML Security Settings page opens.
3. Click the **SAML Enabled** dropdown menu, then select **Enabled**.
4. Under the SAML Enabled dropdown menu, to the far right side of the page, click .
 - » The Add Meta Data Configuration dialog box appears.
5. Enter the required information.

Option	Description
Meta-Data URL	The URL of the identity provider metadata XML file or the location of the downloaded XML file. Important The name of the XML file must begin with the prefix file: and typically uses file:/opt/owl/config/idp-metadata.xml format.
Meta-Data Label	The name for this specific configuration.
IDP URL	The URL of the Collibra DQ application that is provisioned by the identity provider.

Note The IDP_URL is the IdP Initiated SSO login URL. For Example, in Azure, it would be `https://myapps.microsoft.com/signin/<sp-app-id>?tenantId=<sp-tenant-id>`. In multi-tenant mode, this URL should contain a `RelayState` for DQ to set the tenant on completing the SSO. For more information, go to [Configuring multi-tenant support through SAML RelayState](#).

6. Click **Add**.
7. Click the **AD Security Settings** icon in the left navigation page.
 - » The AD Security Settings page opens.
8. Click **Mappings**.
9. Edit the existing tenant admin group in the IdP and assign it the role `ROLE_TENANT_ADMIN`.

10. Restart the DQ Application and test the login.